Reward Hacking Detection via Mechanistic Interpretability in RL Agents

Problem Description

A central challenge in AI alignment is ensuring that reinforcement learning (RL) agents pursue the objectives their designers intend, rather than exploiting incidental features of the reward function. Reward hacking, the phenomenon whereby an agent achieves high reward without genuinely satisfying the intended task, remains one of the most practically dangerous failure modes in deployed RL systems. Recent empirical work shows that such misalignment can remain covert, with agents producing safe-looking outputs while internally pursuing unintended strategies. These findings suggest that behavioral monitoring, which detects hacking after it manifests in outputs, may be insufficient.

This project addresses the following question:

Can we detect reward hacking from the inside of an RL agent, before it manifests in observable behavior?

Rather than relying solely on observable behavior, we shift the focus to the internal representations of the policy itself. The key intuition is that reward-hacking behavior should leave distinct signatures in the model's internal activations, even when external behavior appears aligned. By analyzing these internal states, we aim to detect misaligned strategies at an earlier stage, before they fully manifest in the agent's actions. This perspective reframes reward hacking as a representational phenomenon, where misalignment can be identified through the structure of learned features rather than through outputs alone.

We study this setting in controlled environments with deliberately hackable reward functions, starting with grid-world and, if time permits, extending to continuous-control domains such as MuJoCo.

Related Work

This project sits at the intersection of two active research directions: reward hacking and mechanistic interpretability.

On the reward design side, prior work such as Krakovna et al. (2020), Pan et al. (2022), and Booth et al. (2023) highlights how misspecified reward functions can lead to unintended and often hard-to-detect behaviors. These works emphasize that optimizing proxy rewards does not necessarily imply alignment with the true objective.

On the interpretability side, recent work demonstrates that Sparse Autoencoders can recover interpretable, monosemantic features from neural activations. Bricken et al. (2023) show this for transformer residual streams, and Templeton et al. (2024) demonstrate that this approach scales to frontier models. More directly relevant, Wilhelm et al. (2026) propose internal activation-based monitoring using SAE features to detect reward-hacking behavior in language models, while Zhang et al. (2025) introduce SARM, which integrates SAE representations into reward modeling for feature-level attribution.

While these works focus on large language models, our project aims to extend this interpretability-based detection paradigm to classical RL agents in grid-world and continuous-control domains.

Proposed Method

Our approach consists of the following steps:

- **Design environments with deliberately hackable reward functions:** Grid-world (MVP) where an exploit path such as repeatedly triggering a reward state is available, and MuJoCo environments where shaping rewards can be exploited without genuine task completion.
- **Train RL policy (PPO or SAC):** Train until both genuine-completion and reward-hacking behaviors emerge, and collect labeled trajectories for both behaviors.
- **Extract internal activations:** Record hidden states from the policy network at each timestep.
- **Train Sparse Autoencoder (SAE):** Learn a sparse dictionary of features from activations.
- **Project activations into SAE feature space:** Obtain sparse feature representations for each timestep.
- **Train classifier:** Use episode-level aggregation of features and train logistic regression or a small MLP to predict reward hacking versus genuine completion.
- **Analysis:** Identify which SAE features are most predictive of reward hacking.

Hypothesis

We hypothesize that a classifier trained on SAE-derived sparse features of a policy's internal activations will achieve significantly above-chance detection of reward hacking episodes, with a target AUROC of at least 0.70 in the grid-world setting.

We further hypothesize that this approach will outperform classifiers trained on raw activations due to improved disentanglement, and that the most predictive SAE features will correspond to interpretable behavioral concepts such as proximity to exploit states or divergence from goal-directed trajectories.

Experiment and Evaluation

We will train policies until both hacking and genuine-completion trajectories are reliably obtainable, targeting at least 500 labeled episodes of each class per environment. The dataset will be split into training, validation, and test sets using a 70/15/15 split with stratification by class.

We evaluate classifiers using AUROC, F1-score, and accuracy on the held-out test set. As baselines, we compare against a classifier trained on raw activations, a classifier trained on behavioral features such as trajectory statistics, and random chance.

The key evaluation criterion is whether the SAE-based classifier outperforms both raw-activation and behavioral baselines. For interpretability analysis, we will apply feature attribution methods such as SHAP values or ablation.

Success is defined as achieving an AUROC of at least 0.70 on the test set, outperforming the raw-activation baseline by a statistically significant margin, and identifying predictive features that admit a human-interpretable description consistent with exploitative behavior.